Performance Attribution and Risk Analytics

Version 2 addendum to the eight-ETF portfolio study · retrospective simulation, 2021-2025 · independent project

Question

Version 1 found that the diversified Main portfolio returned 34.81% against 44.68% for a 60% VTI / 40% BND benchmark. This addendum asks **where** that gap came from, **where the risk** sits, and whether standard 1-day VaR models would have described the portfolio's losses out of sample.

Method

Attribution. Brinson-Fachler by calendar month on two segments (equity, fixed income), linked over 60 months with Carino smoothing so effects add exactly to the cumulative arithmetic active return. The model rebalances to target at each month end with no costs, so each month is buy-and-hold and the monthly holdings-based return equals the daily simulation exactly. Selection plus interaction is split by holding as weight x (holding return - benchmark segment return). **Risk.** Euler decomposition of volatility and tracking error at policy weights using the full-period covariance; a trailing 252-day version at each month end. **VaR/ES.** 1-day forecasts at 95% and 99% from a 250-day window using historical simulation (current weights applied to past asset returns), parametric normal, and EWMA (lambda 0.94); backtested on 1005 days from 2021-12-30 with Kupiec and Christoffersen tests. Every forecast uses only data available the prior evening.

Key findings

• Main trailed the 60/40 benchmark by 9.87 pp cumulatively (2021-2025). Brinson-Fachler attribution assigns 0.00 pp to allocation and -9.87 pp to selection: both portfolios start every month at 60% equity / 40% fixed income, so the gap comes from what is held inside each sleeve, not from the stock-bond mix.

• The equity sleeve explains -9.49 pp of the gap. Relative to holding VTI, the US small-cap (-3.41 pp), developed international (-3.53 pp) and emerging-market (-2.55 pp) positions all detracted; the fixed-income sleeve contributed -0.39 pp.

• The diversified mix helped in 2022 and 2025 and hurt in 2021, 2023, 2024; the result is period-dependent and does not by itself show that the diversification choice was wrong.

• For the Conservative portfolio (40/60), allocation explains -16.94 pp of its -23.62 pp gap, showing the attribution separates a lower equity weight from security choice.

• Equity holdings generate 91% of Main's ex-post volatility despite a 60% weight. The 30% VTI underweight accounts for 86% of the 2.27% tracking error.

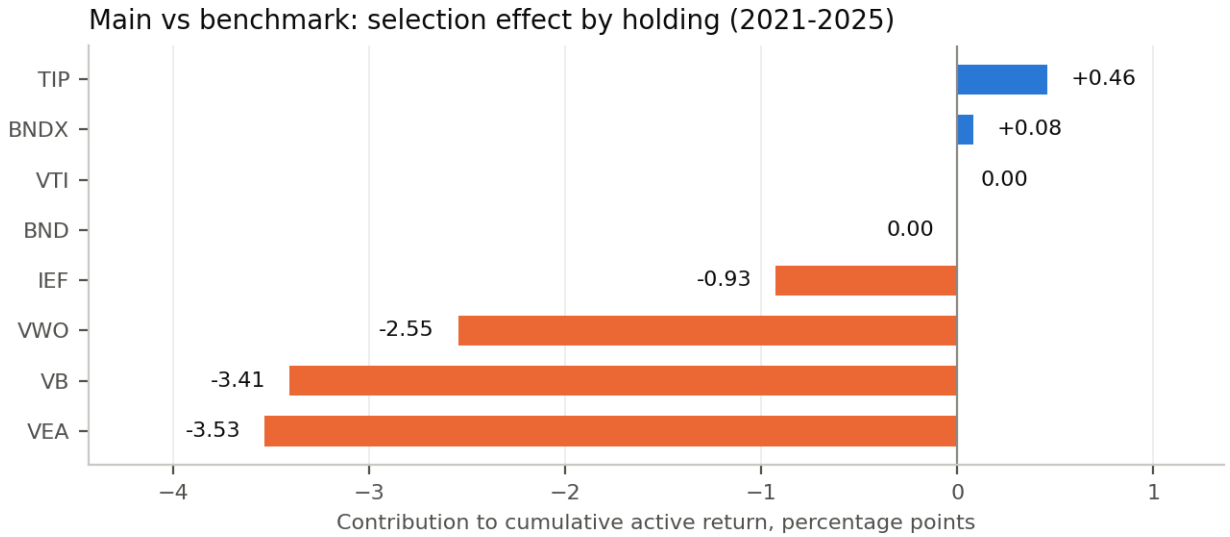
• Fixed income's share of rolling 1-year risk peaked at 20% in Jun 2024, when the VTI-BND correlation was +0.39; bonds diversified less when stock-bond correlation turned positive.

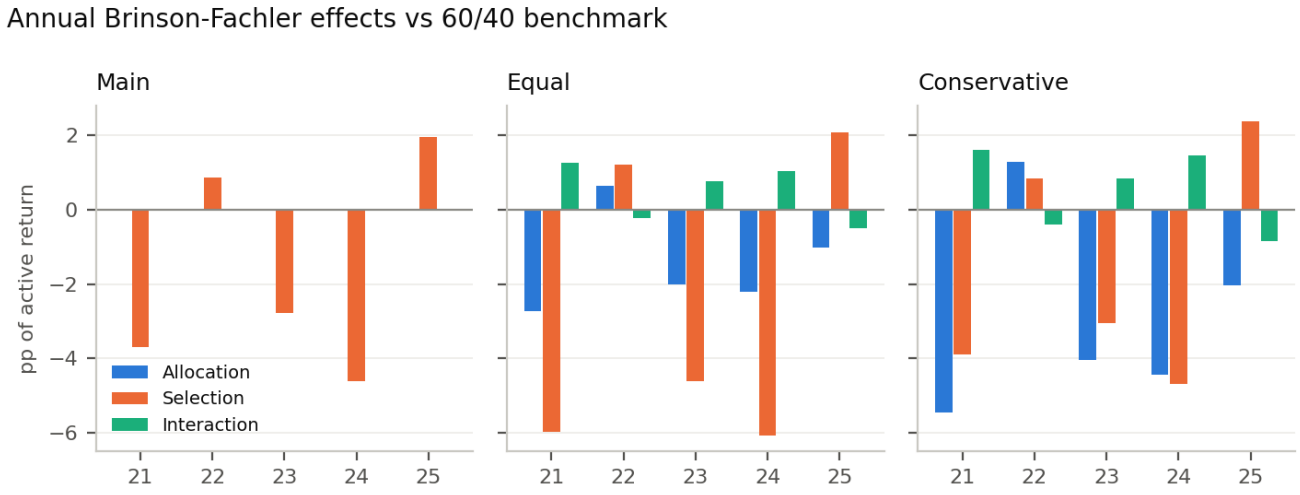
• 1-day 95% VaR is statistically acceptable for all three methods (Kupiec p >= 0.05: yes). At 99%, all methods saw more exceptions than the 10 expected over 1005 days (historical 20, normal 23, EWMA 17); EWMA was closest (Kupiec p = 0.045), consistent with fat tails and volatility clustering.

Attribution results

Portfolio	Return	Benchmark	Active	Allocation	Selection	Interaction
Main	34.81%	44.68%	-9.87 pp	0.00 pp	-9.87 pp	0.00 pp
Equal	23.25%	44.68%	-21.43 pp	-8.52 pp	-15.58 pp	+2.66 pp
Conservative	21.06%	44.68%	-23.62 pp	-16.94 pp	-9.71 pp	+3.03 pp

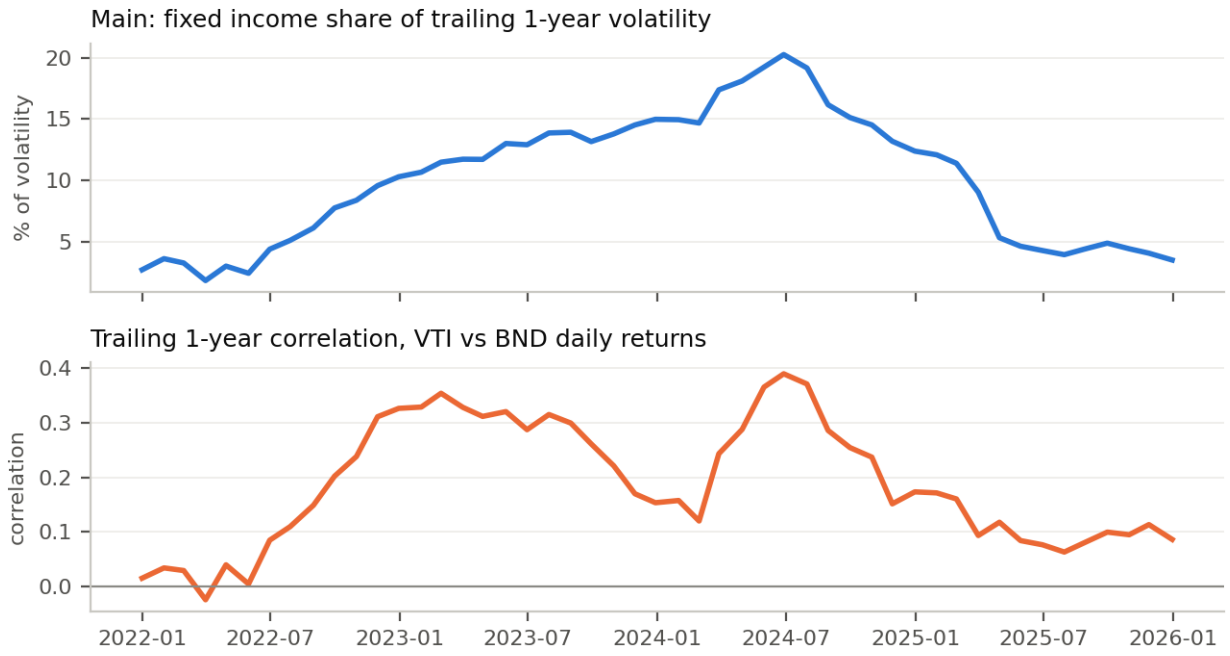
Cumulative 2021-2025, zero-cost scenario. Effects sum exactly to active return (checked to 1e-10).





Risk decomposition (Main, policy weights)

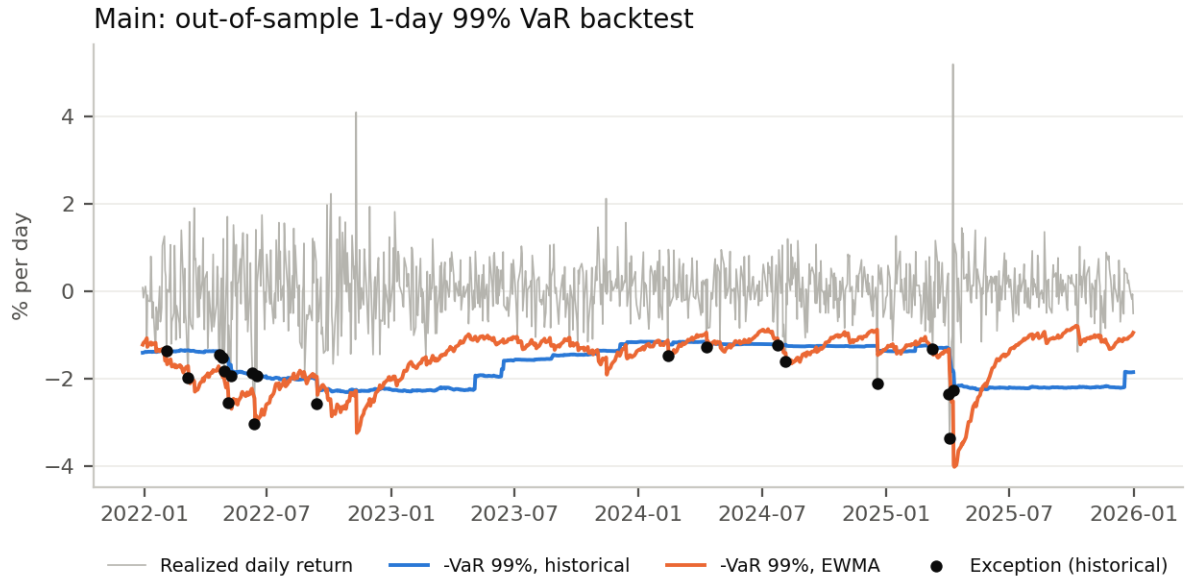
Ticker	Weight	Active wt	% of volatility	% of tracking error
VTI	30%	-30%	46.7%	86.2%
VB	10%	+10%	18.0%	-12.3%
VEA	15%	+15%	20.2%	15.6%
VWO	5%	+5%	6.1%	8.5%
BND	20%	-20%	4.8%	-2.9%
IEF	10%	+10%	2.3%	3.3%
TIP	5%	+5%	1.2%	1.1%
BNDX	5%	+5%	0.8%	0.5%
Total	100%	0%	vol 10.67%	TE 2.27%



VaR backtest (Main)

Method	Conf.	Expected	Exceptions	Kupiec p	Indep. p	Cond. cov. p	Avg VaR	Avg ES
historical	95%	50.3	58	0.273	0.060	0.094	1.05%	1.46%
historical	99%	10.1	20	0.005	0.061	0.004	1.70%	2.04%
normal	95%	50.3	50	0.971	0.012	0.042	1.12%	1.41%
normal	99%	10.1	23	0.000	0.108	0.001	1.59%	1.82%
ewma	95%	50.3	49	0.856	0.785	0.948	1.09%	1.37%
ewma	99%	10.1	17	0.045	0.288	0.076	1.54%	1.76%

p < 0.05 rejects correct coverage (Kupiec), independence of exceptions (Christoffersen) or both (conditional coverage). The same tests for the benchmark are in var_backtest.csv.



Verification

21 unit tests (11 from version 1, 10 new: hand-computed Brinson example, Carino linking, reconciliation with the daily simulation, Euler contributions, Kupiec by hand, clustering detection, VaR window quantile and a no-look-ahead perturbation test). Independent re-derivations in verification_v2.json: status **passed** at tolerance 1e-10.

Limitations

Retrospective, single five-year window chosen in 2026; results are period-dependent. Two-segment attribution only, because the benchmark holds no international or small-cap sleeve; selection is measured against VTI and BND. Attribution covers the zero-cost case. Covariance-based risk uses realized data, not a forecasting risk model; no factor model, stress test or liquidity view. Adjusted prices are a total-return proxy; no taxes, slippage or cash. This is not investment advice.